

Open Source as Institutional Infrastructure

Lessons from the UC OSPO Network

Tim Dennis

Data Science Center, UCLA Library

UCLA OSPO Lead, UC OSPO Network

The invisible layer



Research data workflows run on open source:

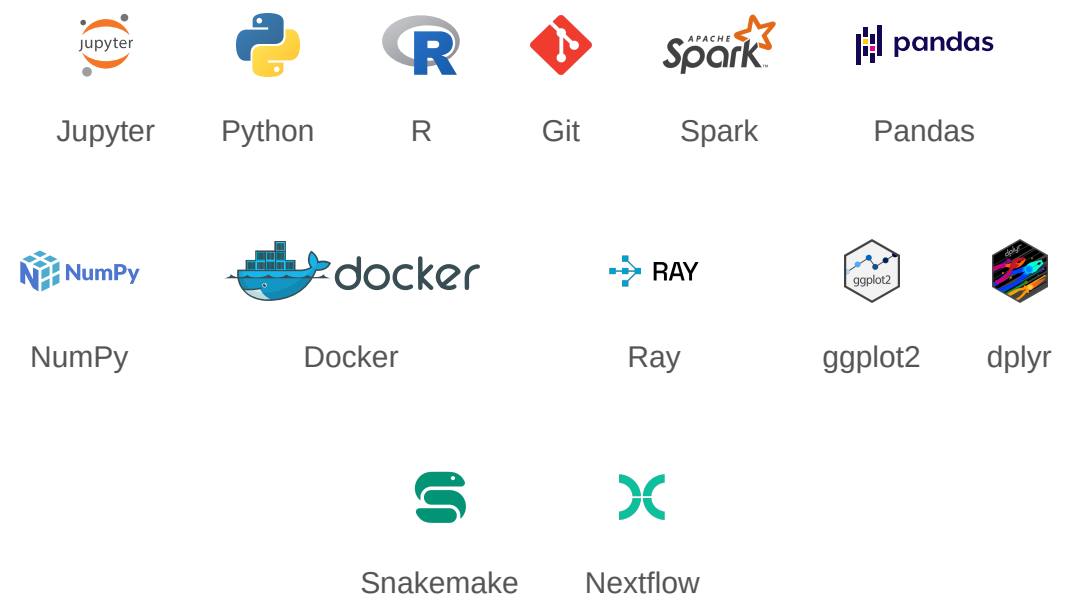
- Data collection, cleaning, analysis, visualization
- Reproducibility tooling: containers, workflows, version control
- Repositories and scholarly infrastructure

92% of researchers use software; **67%** say their research would be impossible without it.

The people keeping it running are increasingly called **Research Software Engineers (RSEs)**, a growing professional community with no standard institutional home.

If every organization had to replace open source with proprietary equivalents, the cost would exceed **\$8.8 trillion**.

Hettrick et al. (SSIJ, 2014/2022) · Hoffmann, Nagle & Zhou (2024)
doi:10.2139/ssrn.4693148



The UC research-to-infrastructure pipeline



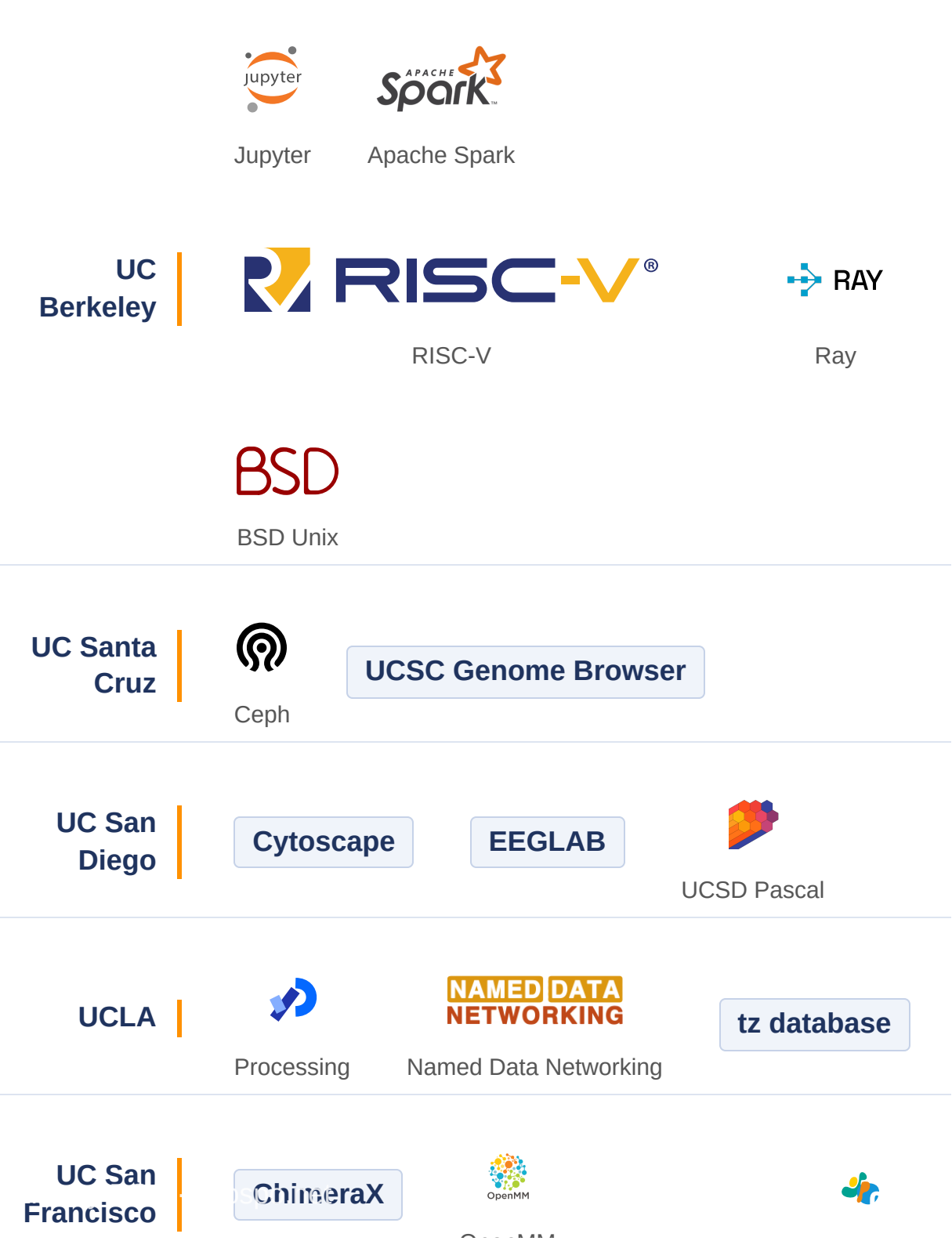
Data professionals built open data infrastructure:
FAIR, DMPs, curation, provenance.

FAIR4RS is the same inflection point for software
— and UC has been showing the way for 50
years.

BSD Unix · Ceph · Apache Spark · Jupyter ·
RISC-V · Ray

The pipeline from university research to global
infrastructure is not rare or accidental.

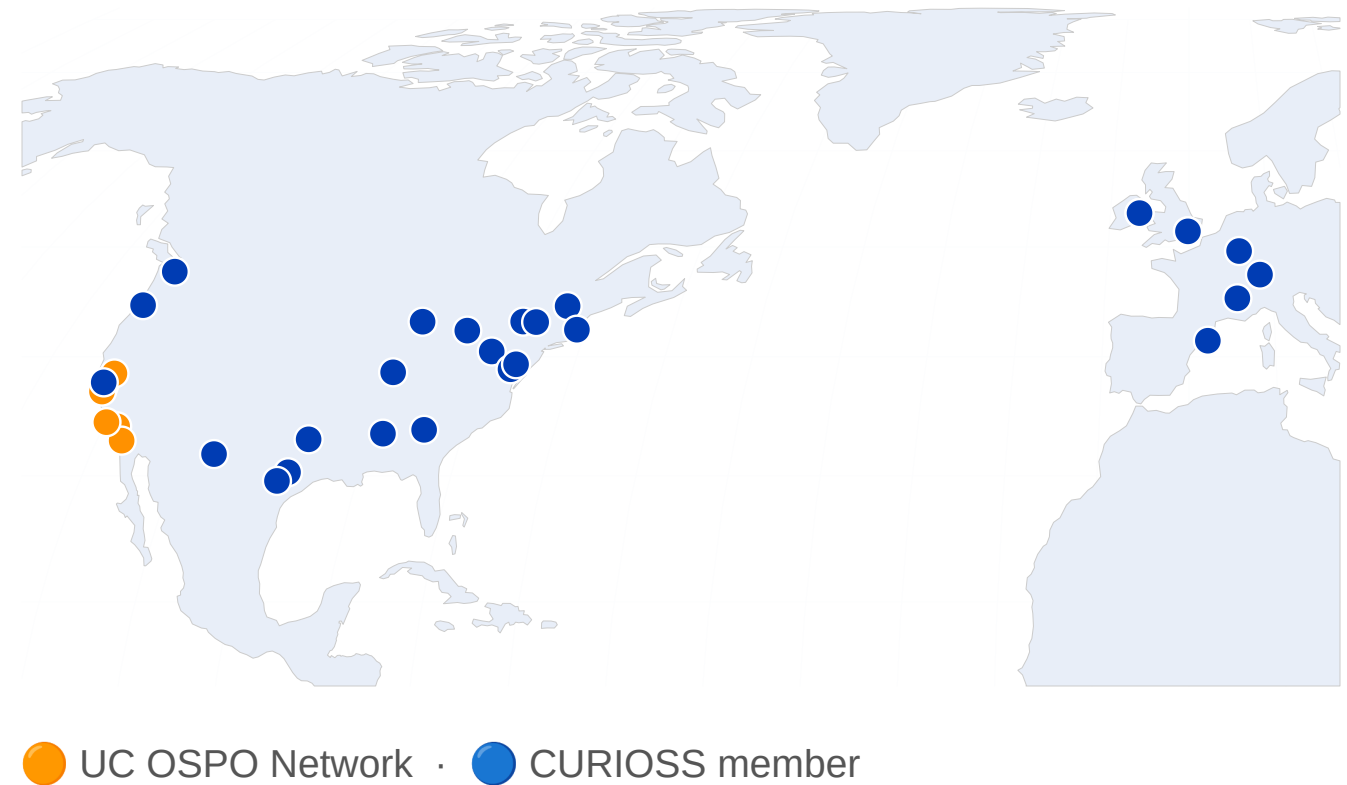
Barker et al. (2022) doi:10.1038/s41597-022-01710-x



What is an academic OSPO?

An **Open Source Program Office** supports, governs, and promotes open source within an institution.

- Originated in industry: ~77% of large tech companies have one
- First academic OSPO: Johns Hopkins, 2019
- **34 academic OSPOs** in CURIOSS worldwide and growing
- Core functions: policy, licensing guidance, best practices, education, community



The UC OSPO Network



Discovery · Sustainability · Education



Discovery

Mapping who does what across the UC system

- 200,000+ repos scanned; ~52,000 institutionally affiliated
- 294-respondent system-wide survey
- UC Open Repository Browser (UC ORB)



Sustainability

Keeping projects and communities healthy

- **58%** of UC contributors are also maintainers
- Cross-campus licensing working group (UCOP + tech transfer)
- Project health assessment (OSSPREY)



Education

Coordinated training across campuses

- Gap analysis → curriculum inventory → learning pathways
- Published at ucospo.net/education
- Coordinated with Carpentries infrastructure



Lessons from two years

What the network model enables that single campuses cannot:

1. **Shared staffing:** community manager, licensing specialist, technical roles
2. **Policy leverage:** the network has standing with UCOP; individual campuses do not
3. **Data at scale:** the GitHub and survey analyses require system-wide scope to mean anything
4. **Equity:** smaller campuses get services they could not build alone
5. **A replicable model:** \$1.85M produced something other state systems can follow

Each campus contributes unique expertise; the network routes it to where it's needed.



Where this meets your work

For data professionals, the OSPO sits at the intersection of:

- Research software engineering ↔ data management
- Licensing compliance ↔ open data policy
- Reproducibility tooling ↔ FAIR / FAIR4RS
- Training infrastructure ↔ data literacy programs
- RSE community ↔ library research support

For your institution:

- Where does open source software governance currently live?
- What would it take to treat it as infrastructure rather than a side project?



You're already doing this work



Learn more / connect

UC OSPO Network: ucospo.net

- Education site: ucospo.net/education
- Mailing list, Slack, events calendar

Global network: curiooss.org, 34 academic OSPOs worldwide and growing

Tim Dennis Data Science Center, UCLA Library
tdennis@library.ucla.edu



Slides: tinyurl.com/iassist2026



Handout: tinyurl.com/iassist2026-handout



QR code linking to tinyurl.com/iassist2026
— slides for this presentation

tinyurl.com/iassist2026

